

ZEAL COLLEGE OF ENGINEERING & RESEARCH, PUNE*(Affiliated to Savitribai Phule Pune University)*

Department of Computer Engineering

FINAL YEAR PROJECT PROPOSAL**Autonomous Enterprise Intelligence Platform
(AI CEO)***A Self-Improving Multi-Agent Operating System for the Mid-Sized Enterprise***Submitted by**

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Under the Guidance of

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1. Introduction

A mid-sized company typically already runs a CRM, an accounting/ERP suite, an HRMS, a support-ticket system, and a project tracker yet almost none of that data talks to any of the rest. A churn-risk signal sitting in the support system never reaches the sales team renegotiating that account; a hiring plan in the HRMS is built without reference to the revenue forecast sitting in the CRM. Large enterprises solve this with expensive, custom-integrated platforms (Salesforce Agentforce, SAP Joule, Microsoft Copilot); small and mid-sized businesses typically cannot justify that cost or complexity.

This project proposes the Autonomous Enterprise Intelligence Platform ("AI CEO"): a multi-agent decision layer that sits above a company's existing systems of record, reads across all of them through a single knowledge graph, and drafts but does not silently execute the cross-functional decisions a human executive would otherwise have to piece together by hand.

2. Problem Statement

Consider a single, common cross-functional event: a top-20 customer files three support tickets in a week. Today, that fact lives only in the support system. Sales does not see it before the renewal call; Finance does not see it before forecasting that account's revenue; Product does not see it as a churn-risk-weighted signal when prioritizing the next sprint. This project targets four such recurring gaps:

- Fragmented customer risk signals churn indicators in one system never reach the teams that need them.
- Manual cross-functional reporting executives spend hours assembling numbers that already exist in five different systems.
- Reactive budget allocation marketing/resourcing budgets are set periodically instead of adjusted against real-time performance.
- No shared institutional memory contracts, policies, and past decisions live in scattered documents no one can query in natural language.

As in plant-floor automation, the underlying problem is not a lack of data but a lack of connection, context, and coordinated reasoning across systems which motivates the same knowledge-graph-plus-agent architecture applied here to enterprise data.

3. Literature Survey / Existing Systems

A brief survey of existing platforms in this space, and the gap this project addresses:

System / Approach	Strength	Limitation this project addresses
Microsoft Copilot (enterprise)	Deep Microsoft 365 / Dynamics integration	Priced and configured for organizations already on the full Microsoft stack
Salesforce Agentforce	Native CRM-object reasoning	Requires an existing Salesforce deployment; licensing scales with seats and data volume
SAP Joule / ServiceNow AI	Deep ERP / workflow integration	Assumes a multi-year, multi-million-dollar enterprise deployment already in place

System / Approach	Strength	Limitation this project addresses
Single-purpose BI dashboards	Good at visualizing one system's data	Cannot reason across systems or propose actions, only display numbers

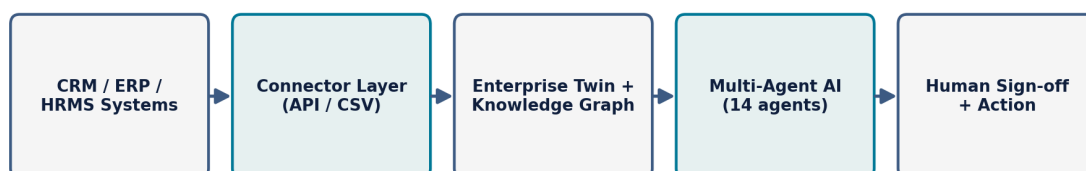
The gap is consistent across all of the above: no accessible platform combines a lightweight knowledge graph, retrieval-augmented reasoning, and a coordinated multi-agent layer that a small or mid-sized company or a student team can stand up without a multi-month integration project. This is the gap this project is scoped to fill.

4. Objectives

- To design lightweight, read-only connectors that ingest CRM/ERP/HRMS-style data (via API or CSV) into a common schema.
- To build an enterprise knowledge graph linking employees, customers, deals, projects, and support tickets, enabling cross-functional queries a single system cannot answer alone.
- To implement a hybrid retrieval pipeline (graph traversal plus vector search) so an Enterprise Copilot can answer questions using both structured facts and unstructured documents.
- To design a multi-agent architecture in which functional agents (finance, HR, sales, marketing, and others) propose recommendations reconciled by a CEO Agent and approved by a human before any external action.
- To build core forecasting models: customer churn prediction and revenue/demand forecasting, each matched to the data volume a mid-sized company actually has.
- To evaluate the system against a public/simulated enterprise dataset, demonstrating measurable improvement over manual cross-functional reporting.

5. Proposed System Architecture

The system mirrors the same five-stage pipeline used for industrial automation, retargeted at enterprise systems of record. Data from CRM/ERP/HRMS systems is ingested through lightweight connectors, folded into an enterprise twin and knowledge graph, reasoned over by a team of specialized agents, and only acted on after a human reviews the CEO Agent's reconciled recommendation.



5.1 Layer Description

Layer	Function
CRM / ERP / HRMS Systems	Existing systems of record already in use by the company (or a demo instance such as Odoo for the prototype)
Connector Layer	Read-only API connectors and CSV ingestion that normalize data from each system into a common schema
Enterprise Twin + Knowledge Graph	Live org-state model plus a Neo4j graph linking employees, customers, deals, and tickets
Multi-Agent AI	Fourteen specialized agents (finance, HR, sales, marketing, legal, and others) coordinated by a CEO Agent
Human Sign-off + Action	An executive reviews the CEO Agent's reconciled recommendation before any external action (e.g. sending a contract) is taken

6. Scope of the Project

6.1 In Scope (Prototype)

- Read-only connectors to a demo CRM/ERP instance (e.g. Odoo demo data) or a public enterprise-style dataset.
- A working knowledge graph, hybrid RAG pipeline, and multi-agent reasoning layer demonstrated end-to-end.
- A churn-prediction model and a revenue-forecasting model, each validated against a held-out baseline.
- An Enterprise Copilot chat interface answering cross-functional questions from the live graph and documents.

6.2 Out of Scope (Future Work)

- Write-access integration with a live production CRM/ERP of a real company.
- Fully autonomous execution of agent recommendations without human approval.
- Support for every possible enterprise system (the prototype targets a representative subset: CRM, accounting, support).

7. Technology Stack

Layer	Technologies
Connectors	REST/GraphQL API connectors, CSV/Excel ingestion via pandas
Storage	PostgreSQL, Neo4j, Milvus/FAISS (vector), Redis
AI / ML	XGBoost/LightGBM (churn), PyTorch Forecasting (Temporal Fusion Transformer), scikit-learn
Generative AI	LLM API for drafts (reports, contracts, emails), retrieval-augmented generation pipeline
Agent Orchestration	LangGraph, CrewAI, Model Context Protocol (MCP)
Workflow Automation	Temporal (durable execution), n8n (lightweight connectors)
Backend	FastAPI, Python
Frontend	React, Recharts

Layer	Technologies
DevOps	Docker, GitHub Actions

8. Methodology and Work Plan

Phase	Duration	Key Deliverable
1. Connector framework	Weeks 1–4	Read-only connectors to demo CRM/ERP data; schema mapping complete
2. Knowledge graph & RAG	Weeks 5–8	Neo4j schema live; hybrid retrieval pipeline over sample documents
3. Forecasting models	Weeks 9–13	Churn model and revenue forecast trained and validated
4. Multi-agent layer	Weeks 14–18	Agent roster implemented; CEO Agent reconciliation working
5. Copilot, integration & testing	Weeks 19–24	End-to-end demo; evaluation report; final documentation

9. Expected Outcomes

- A working prototype demonstrating the full ingest → twin → graph → agents → action pipeline on demo enterprise data.
- A churn-prediction model and a revenue-forecasting model with measurable improvement over a naive baseline.
- A documented, reusable multi-agent architecture shared with the Industrial and Healthcare platform variants developed alongside this project.
- A final project report and, where results warrant it, a workshop-paper draft on agentic AI applications.

10. References

- [1] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” Proc. NeurIPS, 2020.
- [2] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” Proc. ACM SIGKDD, 2016.
- [3] B. Lim et al., “Temporal Fusion Transformers for Interpretable Multi-Horizon Time Series Forecasting,” International Journal of Forecasting, 2021.
- [4] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” Proc. NeurIPS, 2017.

Note to reviewers

This proposal describes the second of three related platform concepts developed by the author (Industrial, Enterprise, and Healthcare variants of the same underlying architecture). The Enterprise platform reuses the agentic-orchestration and knowledge-graph infrastructure built for the Industrial platform, retargeted at enterprise-style data; a longer-form technical design document covering all three platforms in full implementation depth is available on request.